



INTRODUCTION TO CIRCULAR AGRICULTURE

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MODULE OVERVIEW

This module introduces the concept of circular agriculture, focusing on entofarming (insect farming) as a practical example of sustainable food production. It explains how circular systems reduce waste, reuse resources, and recycle nutrients, while helping learners understand the environmental and social importance of sustainable food systems. The module is designed for secondary school teachers and students and encourages critical thinking about future food production and green innovation.

TARGET GROUP

Age Range: 12–18 years

Subject Links: STEM, Biology, Chemistry, Environmental Science, Civic Education, Economics

LEARNING OBJECTIVES

By the end of this module, learners will be able to:

- Explain the concept of circular agriculture and how it differs from traditional linear farming systems.
- Describe the role of entofarming (insect farming) in sustainable food production.
- Identify how circular agriculture reduces waste and improves resource efficiency.
- Recognise the environmental and social benefits of sustainable and circular food systems.



Who are we?

About the Project

This module has been developed within the framework of an **Erasmus+ proposal** focusing on **sustainable agriculture, circular economy, and innovative learning** approaches in STEM education.

The project brings together educational institutions, universities, and organizations from **Greece, Bulgaria and North Macedonia** to develop engaging educational materials for secondary school students. Its main objective is to help young people better understand how sustainable practices can address environmental challenges while **encouraging interest in STEM education and career**.

A central element of the project is the EntoFARM educational concept, which introduces students to alternative and sustainable food production systems such as insect farming and other circular resource management practices. Through these examples, students explore how biological processes, technological innovation, and responsible resource use can contribute to more sustainable food systems.

The educational materials produced in the project aim to support teachers in introducing sustainability and circular economy concepts through interactive and interdisciplinary learning activities.

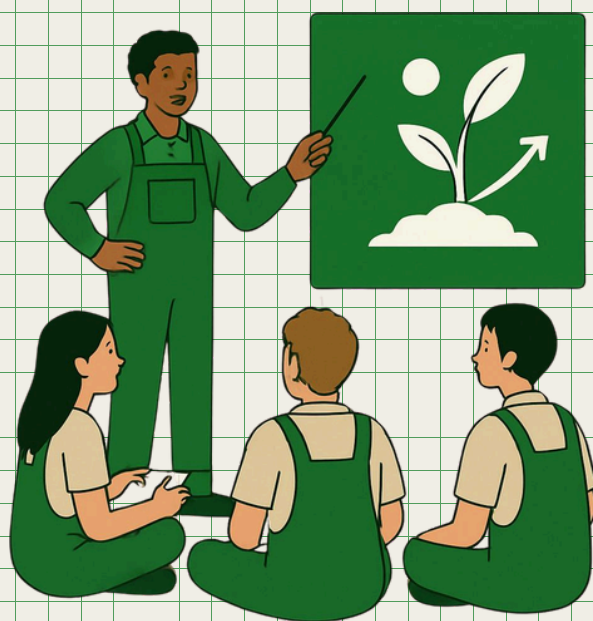


Project Consortium:

- Fraud Line – Greece (Project Coordinator)
- University of Thessaly – Greece
- University of Patras – Greece
- 1st EPAL Volou – Greece
- Professional Agricultural School “Buzema” – Bulgaria
- Professional Forum for Education – Bulgaria
- Association for Internationalization of Education and Science – North Macedonia

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WHY EDUCATE STUDENTS ABOUT CIRCULAR AGRICULTURE?



Education about circular agriculture and food systems creates valuable learning opportunities for students. Young people are naturally curious about food—where it comes from, how it is produced, and how it affects the environment. Circular agriculture provides answers to these questions while encouraging critical thinking and problem-solving.

However, circular agriculture is not yet recognised as a separate subject in many secondary school curricula, including those of the EntoFarm project partners. As a result, many teachers may not feel fully prepared to address student questions in this area. When students or teachers search for information online, they may encounter conflicting or misleading sources. Because circular agriculture is complex, understanding it requires learning about science, ecology, economics, and social responsibility. Schools, the agri-food sector, and circular agriculture initiatives all share a responsibility to provide accurate and balanced information. Without proper education, myths about farming and food systems can spread and slow progress toward sustainability.

Learning about circular agriculture, especially through practical examples like entofarming helps students:

- Understand how waste can become a resource.
- See the connection between food production and environmental protection.
- Develop interest in local and sustainable food systems.
- Explore future careers in agriculture, science, and green innovation.

For example, biology lessons can be linked to insect life cycles, nutrition, and ecosystems, while chemistry and environmental science can explore nutrient recycling and waste reduction. These real-world applications help students see how classroom learning connects to future study and employment opportunities.

CORE PRINCIPLES OF CIRCULAR AGRICULTURE: FROM LINEAR TO CIRCULAR FOOD SYSTEMS



Circular agriculture represents a major shift away from the traditional linear farming model of “take – produce – consume – discard”.

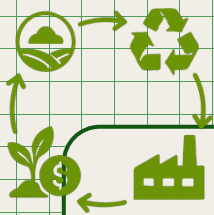
In linear systems, resources are often used once and then wasted. Circular agriculture replaces this approach with a regenerative and sustainable system, where materials are kept in use for as long as possible. The main goals of circular agriculture are to:

- Avoid waste and losses.
- Reuse and recycle by-products.
- Use nutrients and biomass efficiently.
- Protect natural ecosystems while producing food.

A strong example of this approach is entofarming. Insect farming allows food waste, crop residues, or by-products from the food industry to be transformed into valuable outputs such as:

- High-quality protein for animal feed or human food.
- Organic fertilizer (frass) that improves soil health.
- Reduced need for imported feed and chemical inputs.

By adopting circular practices like entofarming, farming systems become more resilient, cost-effective, and environmentally friendly. This helps farmers improve food security, reduce dependence on external resources, and build self-sustaining production systems.



Circular business models in agriculture

Circular agriculture is closely linked to circular business models, which aim to keep resources in use for longer and recover value at every stage of production. These models can be grouped into four main types:

CORE PRINCIPLES OF THE CIRCULAR ECONOMY

1. **Product-based models:**

These focus on producing high-quality, long-lasting products. In agriculture, this includes durable farming equipment or well-designed insect-rearing systems that reduce waste and costs over time.

2. **Service-based models:**

In this model, companies keep ownership of products and provide services instead. For example, a company might manage insect farming units or waste-processing services for farmers or food businesses.

3. **Sharing-based models:**

These models increase efficiency by sharing resources, such as farming equipment, processing facilities, or digital platforms for knowledge exchange.

4. **Supply chain-based models:**

These reduce the need for new raw materials by recovering energy and nutrients from by-products and waste. Entofarming fits perfectly here, as insects convert organic waste into valuable feed and fertilizer.

Together, these models support closed-loop or circular agriculture, where “nothing is wasted” and every output has a purpose.

Why circular agriculture matters for the future:

Circular agriculture offers a powerful framework for transforming modern farming into a system that works with nature, not against it. It:

- Reduces pressure on natural resources.
- Helps fight climate change.
- Supports biodiversity.
- Improves long-term food security.

Global food demand is expected to grow significantly in the coming decades. At the same time, climate change and biodiversity loss threaten the food systems we all depend on. Changing how we produce food by following circular economy principles is one of the most effective ways to address these challenges. Farmers play a key role in this transition. By choosing regenerative and circular farming strategies, including insect farming, they can:

- Improve soil health.
- Use resources more efficiently.
- Reduce greenhouse gas emissions.
- Create economic and social benefits for society.

Regenerative circular practices also help farmers reduce costs, improve yields, and become more resilient to extreme weather and market changes. New opportunities may even arise, such as earning income through environmental services like carbon storage in soils.

CORE PRINCIPLES OF THE CIRCULAR ECONOMY

Circular agriculture across the food system

In circular agriculture, every step of the food system is designed with sustainability in mind:

- Growing and harvesting.
- Processing and packaging.
- Transport and marketing.
- Consumption and waste management.

Insect farming can be integrated at several points in this system, especially in waste reduction, feed production, and soil regeneration. This makes entofarming an excellent teaching example of how circular principles work in real life. Farmers are also business decision-makers who respond to market demands. By developing green skills, leadership, and knowledge of circular systems, more farmers can be supported and motivated to adopt sustainable practices.

CASE STUDY: TURNING FOOD WASTE INTO PROTEIN WITH ENTOFARMING

Background:

In many cities, large amounts of food waste are produced every day by households, schools, and supermarkets. At the same time, farmers need large quantities of protein-rich feed for animals such as chicken and fish. Producing this feed often requires imported soy, which can contribute to deforestation and high greenhouse gas emissions. To address this problem, a small insect farm can be set up near an urban area using black soldier fly larvae as part of a circular agriculture system.



Scenario Summary

Instead of sending food waste to landfill, the insect farm collects safe, pre-sorted food by-products from local sources such as vegetable markets and school canteens. These by-products are used to feed the insect larvae.

The system works as a closed loop:

1. Food waste is collected and prepared.
2. Insects (larvae) feed on the waste and grow quickly.
3. The larvae are harvested and processed into high-protein animal feed.
4. The remaining material (called frass) is used as a natural fertilizer for crops.
5. Crops help produce more food—some of which may again enter the system as by-products.



Purpose:

The purpose of this system is to reduce food waste by converting it into valuable resources instead of sending it to landfill. It supports sustainable agriculture by producing high-protein animal feed and natural fertilizer. Overall, it promotes a circular economy where waste is reused efficiently to create new food and resources.



Reflection questions:

- How does this insect farm reduce waste compared to a traditional food system?
- Why are insects considered efficient protein producers?
- Can you think of places in your community where entofarming could be used?
- What challenges might farmers or communities face when starting an insect farm?

ADDITIONAL RESOURCES

Links:

- Atinkut, H.B.; Yan, T.; Zhang, F.; Qin, S.; Gai, H.; Liu, Q. Cognition of Agriculture Waste and Payments for a Circular Agriculture Model in Central China. *Sci. Rep.* 2020, 10, 10826.
- Burns, E. A. (2021). Regenerative agriculture : farmer motivation, environment and climate improvement. *Policy Quarterly*, 17(3), 54–60. <https://doi.org/10.26686/pq.v17i3.7133>
- Corvellec H., Stowell A., F., Johansson N. (2022). Critiques of the circular economy. *Journal of Industrial Ecology* 2022;26:421–432
- El Janati, M.; Akkal-Corfini, N.; Bouaziz, A.; Oukarroum, A.; Robin, P.; Sabri, A.; Chikhaoui, M.; Thomas, Z. Benefits of Circular Agriculture for Cropping Systems and Soil Fertility in Oases. *Sustainability* 2021, 13, 4713. <https://doi.org/10.3390/su13094713>
- Jaeger-Erben M., Jensen C., Hofmann F., Zwiers, J. (2021). There is no sustainable circular economy without a circular society. *Resources Conservation and Recycling* 168(2):105476, DOI:10.1016/j.resconrec.2021.105476
- Kirchherr J., Reike D., Hekkert M., (2017). Conceptualizing the circular economy: an analysis of 114 definitions *Resour. Conserv. Recycl.* 127 (2017)
- Koul, B., Yakoob, M., & Shah, M. P. (2022). Agricultural waste management strategies for environmental sustainability. *Environmental Research*
- Padilla-Rivera, A., Russo-Garrido, S., Merveille, N., (2020). Addressing the Social Aspects of a Circular Economy: A Systematic Literature Review. *Sustainability*
- Piscicelli, L., & Ludden, G. (2016). The potential of Design for Behaviour Change to foster the transition to a circular economy

TEACHER REFLECTION AND ADAPTATION NOTES

This module has been designed as a flexible educational resource that can be adapted to different classroom contexts, student backgrounds, and educational levels.

Teachers are encouraged to reflect on how the activities and content can be adjusted to best support the needs of their students.

Reflection Questions for Teachers

Before or after implementing the module, teachers may consider the following questions:

- Did the activities engage students with different interests and learning styles?
- Were all students encouraged to participate actively in discussions and group work?
- Did the module help students understand the connection between science, technology, and sustainability?
- Were examples and explanations accessible to students with different levels of prior knowledge?

Accessibility Considerations

To ensure that the module is accessible to all learners, teachers may consider:

- simplifying technical terminology when necessary
- providing visual aids, diagrams, or short videos
- allowing students to express their understanding through different formats (presentations, posters, discussions, or written reflections)
- offering additional guidance or scaffolding for students who may need more support

Supporting Girls' Engagement in STEM

Research shows that girls are more likely to engage with STEM when learning environments are collaborative and when scientific topics are linked to real-world challenges and social impact.

Teachers can therefore:

- encourage mixed-gender teamwork and equal participation
- highlight female role models in sustainability science and agriculture
- frame STEM challenges as problem-solving activities that benefit society and the environment
- ensure that classroom discussions give equal voice to all students

TEACHER REFLECTION AND ADAPTATION NOTES

Adapting the Module

Teachers are free to adapt the module by:

- adjusting the depth of scientific explanations
- modifying activities to fit available time or resources
- connecting the topic to local environmental or agricultural examples
- integrating the module with subjects such as biology, geography, environmental science, or technology

The goal is to ensure that the module remains relevant, inclusive, and meaningful for students in different educational settings.

Do you have any questions?

Please let us know! We are happy to help.

Contact us at:



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